

UE23CS320B-- Capstone Project Phase – II - Weekly Status Report

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Need Approval for Capstone PPT and Weekly Report inbox

P POOJARY SAIRAJ SHANKAR PESU-EC-B.Tech 2023 <pes2ug23am073@pesu.pes.edu>
to Kailasam

6:52 PM (2 minutes ago) ☆ 😊 ↶ ⋮

2 Attachments • Scanned by Gmail • Add all to Drive

 

M Mr. Kailasam P K PESU-EC-ECE
to me

6:54 PM (0 minutes ago) ☆ 😊 ↶ ⋮

Approved

Regards
Kailasam

On Fri, 10 Apr 2023 at 18:52, POOJARY SAIRAJ SHANKAR PESU-EC-B.Tech 2023 <pes2ug23am073@pesu.pes.edu> wrote:

↶ Reply ↷ Forward 🗑

UE23CS320B-- Capstone Project Phase – II - Weekly Status Report

Project Title		Driver Behavior Anomaly Detection using Deep Learning	
Batch No		06	
Week Starting Date		17-02-2026	Week Ending Date 23-02-2026
Name		Chetan Nadichagi Poojary Sairaj Shankar Ranjith Kumar.V Mahesh E Ajjer	SRN PES2UG23CS149 PES2UG23AM073 PES2UG23AM082 PES2UG23CS316
Activities Completed			
#	Tasks	Date	Remarks
1	Established core OpenCV webcam capture pipeline		Enabled real-time frame extraction
2	Implemented BGR to RGB color conversion logic		Standardized input for model accuracy
In Progress			
#	Tasks	Date	Remarks
1	Integrating MediaPipe Face Mesh		Setting up 478-point landmark tracking.
2	Dataset balancing with hockey negatives.		Establishing baseline measurements.
Plan for Next Week			
#	Tasks	Date	Remarks
1	Calculate Eye Aspect Ratio (EAR)		Core feature for micro-sleep detection.
2	Calculate Mouth Aspect Ratio (MAR)		Core feature for yawning detection



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Prof.Kailsam

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Coordinator

Chairperson

UE23CS320B-- Capstone Project Phase - II - Weekly Status Report

Project Title	Real-time AI-Driven Anomaly Detection in Surveillance Cameras		
Batch No	06		
Week Starting Date	24-02-2026	Week Ending Date	02-03-2026
Name	Chetan Nadichagi Poojary Sairaj Shankar Ranjith Kumar.V Mahesh E Ajjer	SRN	PES2UG23CS149 PES2UG23AM073 PES2UG23AM082 PES2UG23CS316
Activities Completed			
#	Tasks	Date	Remarks
1	Successfully integrated MediaPipe Face Mesh.		Extracting precise geometric features
2	Finalized EAR, MAR, and Head Pose logic		Basic rule-based classification achieved
In Progress			
#	Tasks	Date	Remarks
1	Object detection module setup		Transitioning to YOLOv8 architecture.
2	Optimizing frame processing speed		Ensuring real-time performance.
Plan for Next Week			
#	Tasks	Date	Remarks
1	Implement YOLOv8 for mobile phones		Enable manual distraction detection.
2	Combine spatial and object features		Prepare data for temporal sequence

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UE23CS320B-- Capstone Project Phase – II - Weekly Status Report

Project Title		Real-time AI-Driven Anomaly Detection in Surveillance Cameras	
Batch No		06	
Week Starting Date		03-03-2026	Week Ending Date 09-03-2026
Name		Chetan Nadichagi Poojary Sairaj Shankar Ranjith Kumar.V Mahesh E Ajjer	SRN PES2UG23CS149 PES2UG23AM073 PES2UG23AM082 PES2UG23CS316
Activities Completed			
#	Tasks	Date	Remarks
1	Implemented YOLOv8 object detection model.		Bounding boxes for mobile phones active
2	Fused MediaPipe and YOLOv8 data streams		Unified 15-dimensional feature output
In Progress			
#	Tasks	Date	Remarks
1	Designing temporal sequence architecture		Mapping out Bidirectional LSTM (BiLSTM).
2	PyTorch environment configuration.		Setting up deep learning backend.
Plan for Next Week			
#	Tasks	Date	Remarks
1	Build the foundational BiLSTM network		Establish sequence memory learning
2	Generate synthetic training data		Required for initial architecture testing



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UE23CS320B-- Capstone Project Phase - II - Weekly Status Report

Project Title		Real-time AI-Driven Anomaly Detection in Surveillance Cameras	
Batch No		06	
Week Starting Date	10-03-2026	Week Ending Date	16-03-2026
Name	Chetan Nadichagi Poojary Sairaj Shankar Ranjith Kumar.V Mahesh E Ajjer	SRN	PES2UG23CS149 PES2UG23AM073 PES2UG23AM082 PES2UG23CS316
Activities Completed			
#	Tasks	Date	Remarks
1	Generated synthetic (30, 15) sequence data		Simulated real-time driver metrics.
2	Constructed PyTorch BiLSTM neural network		Forward and backward data flow verified.
In Progress			
#	Tasks	Date	Remarks
1	Implementing custom Focal Loss function.		Handling imbalanced behavior data.
2	Designing neural Attention mechanism		Focusing model on critical transitions.
Plan for Next Week			
#	Tasks	Date	Remarks
1	Integrate Attention layer into BiLSTM		Enhance predictive accuracy
2	Train sequence model on synthetic data.		Validate deep learning pipeline logic



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Project Title	Real-time AI-Driven Anomaly Detection in Surveillance Cameras		
Batch No	06		
Week Starting Date	17-03-2026	Week Ending Date	23-03-2026
Name	Chetan Nadichagi Poojary Sairaj Shankar Ranjith Kumar.V Mahesh E Ajjer	SRN	PES2UG23CS149 PES2UG23AM073 PES2UG23AM082 PES2UG23CS316
Activities Completed			
#	Tasks	Date	Remarks
1	Integrated Attention layer and Focal Loss		Completed advanced model architecture.
2	Trained PyTorch model on synthetic validation set		Achieved stable convergence (100%).
In Progress			
#	Tasks	Date	Remarks
1	Developing alert generation module		Creating visual and auditory warnings
2	End-to-end pipeline integration script		Connecting all modules into main.py
Plan for Next Week			
#	Tasks	Date	Remarks
1	Finalize alert system logic		Ensure instantaneous trigger on anomaly.
2	Complete Phase 2 full system testing		Prepare codebase for Review 2 presentation

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Project Title	Real-time AI-Driven Anomaly Detection in Surveillance Cameras		
Batch No	06		
Week Starting Date	24-03-2026	Week Ending Date	30-03-2026
Name	Chetan Nadichagi Poojary Sairaj Shankar Ranjith Kumar.V Mahesh E Ajjer	SRN	PES2UG23CS149 PES2UG23AM073 PES2UG23AM082 PES2UG23CS316
Activities Completed			
#	Tasks	Date	Remarks
1	Finalized multi-modal alert system		Flashing borders and audio beeps active.
2	Successfully ran 30% implementation end-to-end.		Pipeline functioning seamlessly on webcam.
In Progress			
#	Tasks	Date	Remarks
1	Finalizing documentation and Phase 2 PPT		Structuring presentation flow and novelty
2	Reviewing State Farm Distracted Driver dataset		Planning transition to realworld data
Plan for Next Week			
#	Tasks	Date	Remarks
1	Begin downloading/cleaning State Farm dataset		Prepare for heavy phase 3 training
2	Implement data augmentation for real videos.		Improve model robustness to lighting

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